

GUSTAVSFORS, Sweden

WMO: 024260

Lat: 60.151N

Lon: 13.797E

Elev: 617

StdP: 14.37

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-6.3	-16.7	2.3	-12.1	-10.3	3.2	-6.4	16.6	35.9	14.5	35.3	0.1	10	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	79.4	62.9	76.2	61.8	72.9	60.4	66.0	75.1	64.1	72.0	62.4	69.6	4.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	87.8	68.4	61.2	82.5	67.2	59.3	77.1	65.3	31.0	75.1	29.6	71.9	28.4	69.8	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.8	9.3	DB	-17.2	83.3	6.8	2.9	-22.1	85.5	-26.1	87.2	-29.9	88.8	-34.9	91.0	(4)
(5)			WB	-17.5	68.1	6.5	1.9	-22.2	69.5	-26.0	70.6	-29.6	71.7	-34.3	73.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	20.3	22.0	28.7	38.7	47.6	55.2	59.6	56.7	48.9	40.0	31.9	23.1		
	DBStd	15.78	11.97	9.70	8.97	5.96	6.79	5.24	5.04	4.85	5.62	6.75	8.12	12.36		
	HDD50	4630	919	784	661	343	128	12	0	6	85	316	542	834		
	HDD65	9330	1384	1204	1126	790	539	296	180	260	484	776	992	1299		
	CDD50	794	0	0	0	3	54	168	299	214	51	5	0	0		
	CDD65	18	0	0	0	0	0	2	13	3	0	0	0	0		
	CDH74	515	0	0	0	0	27	126	303	59	0	0	0	0		
	CDH80	68	0	0	0	0	1	14	51	2	0	0	0	0		
	WSAvg	3.4	3.1	3.0	3.3	3.5	4.0	3.7	3.3	3.3	3.4	3.4	3.3	3.2		
	PrecAvg	29.0	2.1	1.5	1.4	1.7	2.3	2.9	3.3	3.3	2.6	2.9	2.6	2.3		
	PrecMax	38.7	4.3	3.0	2.8	4.8	5.6	5.9	8.8	5.7	4.4	7.0	5.6	4.4		
(15) Precipitation	PrecMin	23.1	0.4	0.3	0.1	0.5	0.6	0.4	0.9	1.3	0.2	0.9	1.0	0.5		
	PrecStd	3.5	1.1	0.7	0.8	1.0	1.2	1.3	1.5	1.3	1.2	1.4	1.2	1.0		
	DB	43.1	43.8	56.3	66.9	77.7	82.1	84.6	79.0	70.3	59.7	51.6	48.3			
	MCWB	41.2	39.2	45.4	51.4	58.9	62.4	66.0	64.7	60.2	53.9	50.7	46.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	39.8	40.2	49.9	62.1	72.8	77.6	81.0	75.6	66.1	56.0	48.6	44.4			
	2% MCWB	38.2	37.7	41.8	48.8	56.6	61.2	64.4	63.6	57.9	52.5	47.1	43.2			
	5% DB	37.4	37.7	45.0	57.5	68.5	74.0	77.5	72.3	62.9	53.7	45.9	41.5			
	5% MCWB	36.3	36.0	38.4	46.2	54.8	59.6	62.9	62.1	56.3	51.0	44.7	40.3			
	10% DB	35.0	35.3	40.9	53.1	63.3	69.7	73.6	68.7	60.1	51.4	43.5	38.3			
	10% MCWB	33.9	34.0	36.2	43.6	52.2	57.1	61.6	60.6	54.8	49.2	42.4	37.3			
	0.4% WB	41.4	41.1	46.0	53.5	61.7	65.6	68.8	67.3	62.0	55.8	50.7	47.1			
	0.4% MCDB	42.4	42.3	54.1	64.9	72.3	75.1	78.4	74.3	66.9	58.1	51.1	48.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	38.4	38.3	42.3	50.0	59.1	63.0	66.9	65.1	59.4	53.5	47.5	43.3			
	2% MCDB	39.3	39.6	48.8	59.4	69.0	73.7	76.9	71.6	64.2	55.0	48.3	44.2			
	5% WB	36.4	36.1	39.5	47.2	56.4	60.9	65.1	63.3	57.4	51.5	45.0	40.2			
	5% MCDB	37.3	37.2	43.3	55.9	64.9	70.8	74.4	69.4	61.3	53.3	45.8	41.0			
	10% WB	34.1	33.9	37.2	44.8	53.7	58.8	63.2	61.7	55.5	49.5	42.5	37.2			
	10% MCDB	34.9	35.3	40.1	51.5	61.8	68.2	70.7	67.5	59.3	51.0	43.4	38.2			
	MDBR	13.3	16.1	19.8	22.2	22.5	22.5	21.7	20.8	19.6	14.7	10.9	12.2			
	5% DB	11.6	15.8	25.2	32.4	32.5	32.5	30.0	27.6	24.5	17.8	12.8	12.8			
(36) Mean Daily Temperature Range	5% DB	10.9	13.1	17.1	19.9	18.0	17.2	15.2	16.4	16.8	14.1	11.8	12.5			
	5% WB	11.6	13.0	21.9	28.7	26.4	27.1	25.9	23.5	22.2	14.8	12.1	12.6			
	MCWBR	11.1	11.2	15.5	18.3	15.6	15.1	14.2	14.9	16.3	12.6	11.4	12.5			
	MCWBR	11.1	11.2	15.5	18.3	15.6	15.1	14.2	14.9	16.3	12.6	11.4	12.5			
(40) Clear-Sky Solar Irradiance	taub	0.254	0.297	0.324	0.362	0.362	0.353	0.373	0.364	0.333	0.320	0.276	0.231			
	taud	2.200	2.288	2.349	2.352	2.413	2.462	2.422	2.442	2.519	2.460	2.280	2.112			
	Ebn at Noon	168	224	256	266	274	279	269	263	254	216	157	125			
	Edh at Noon	13	21	27	32	33	32	32	30	23	18	12	9			
(44) All-Sky Solar Radiation	RadAvg	91	291	772	1216	1523	1739	1561	1223	812	363	115	58			
	RadStd	15	56	86	151	150	110	204	129	102	69	31	7			

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page